VF-C04 Martindale Abrasion

Test Method for Martindale Abrasion	Test method # VF- C04 REF #: TM-R- 023
Date Issued: Dec 2, 2024 Revision #: 3	Reason for Revision: Define evaluation area.

PURPOSE AND SCOPE: To measure the abrasion resistance of lining materials by passing the test sample in a cyclical pattern over the surface of a wool fabric abradant.

REFERENCE: SATRA TM 31

EQUIPMENT AND MATERIAL:

1. Martindale abrasion machine, (SATRA STM105 or any machine meeting the requirement of SATRA TM31).
2. Fabric abradant, as described in SATRA TM 31A
3. Felt backing material, as described in SATRA TM 31A
4. 2.5kg disc weight
5. PU foam with a thickness of 3±1 mm, density of 30±1 kg/m_3, white or pastel color.
6. Distilled Water
7. Immersion dish.
8. Woven, non-woven, and knitted textile fabrics.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Die or hand cut test sample to a diameter of 44+/-1mm or 38+/-1mm samples should be taken from varying locations on the test material, and at least 10cm from any edge.

EQUIPMENT PREPARATION:

1. Remove the four test sample holders and top plate from the Martindale.
2. Remove the clamping rings (which hold the fabric abradant and felt backing materials in position) from each of the four testing stations.
3. Examine the condition of the felt backing materials, Felt backing (165mm in diameter) requires replacement whenever the surface of the material has been damaged of soiled (backings showing only soiling can be continued to be used by placing the soiled side down on the test station surface).Also, if a felt backing has been used for a wet abrasion test, it should not be used for subsequent dry abrasion tests but it can be used in subsequent wet abrasion tests provided it has not been damaged.
4. Examine the condition of the fabric abradant, fabric abradant (165mm in diameter) should be replaced after each test.
5. The fabric abradant should be placed on the test station such that the finished (smoother) side is facing upward.
6. Place the 2.5kg disc weight on top of the fabric abradant and reposition the clamping ring onto the test station.
7. Reposition the top plate onto the machine.

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REPETITIONS: Test two (2) “dry” samples and two (2) “wet” samples per material.

TEST PROCEDURE:

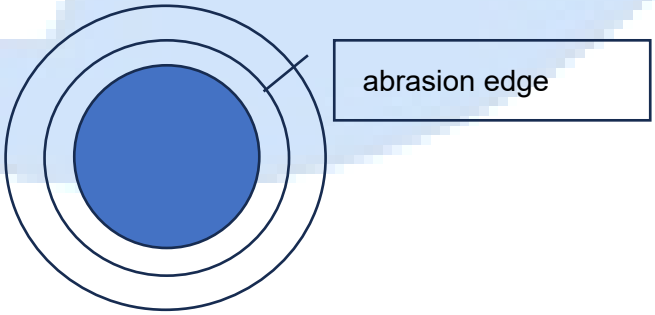
Test should be conducted at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

A. Dry abrasion test

1. Position the test sample into the bottom of the test sample holder such that it is centered and the outer (test) surface faces the hole. Always test the “face” side of the fabric, unless otherwise indicated.
   Note: If the thickness of the test specimen is below 2.5 mm, then place a piece of the PU foam with a thickness of 3±1 mm, and density of 30±1 kg/m3 as backing for test specimen. If the thickness of the test sample is above 2.5 mm, no need for the foam backer.
2. Place the sample weight /locator on top of the test sample such that it is also centered in the holder.
3. Screw the ring clamp into the holder such that the test fabric remains centered in the holder and free of wrinkles, it is important that the ring clamp be firmly screwed onto the test fabric, such that the fabric does not slip in the holder during testing.
4. Place the assembled sample holder face down onto the fabric abradant test station.
5. Insert the spindle /weight through the opening in the top plate and into the opening in the sample holder, a loading of 12 kPa is always used. Check that the spindle moves freely in the opening of the top plate.
6. Set the counter to zero, set the pre-set counter to the desired number of cycles, and start the machine.
7. Examine the samples at least every 2,000 cycles.
8. Discontinue the test when the first yarn has been completely worn through, prior to this failure point, it is also important to record any relevant qualitative observations concerning the condition of the test fabric (i.e. pilling, color change, or any other noticeable change in the appearance of the test fabric), and the number of cycles at which the observation was made.

Note: If the appearance of the test material has been damaged to a point which is viewed as unacceptable for use on a shoe (but prior to the complete wear through of a yarn) the test material should be failed at this point and testing discontinued.

9. The dry test should be run for a maximum of 50,000 cycles, if no damage has occurred at this point, discontinue the testing, and record the result as “50,000+cycles).
10. For specimen evaluation, the damage of the specimen must be checked in the center of the abraded area and not at the edges. **No need** record damage or breakage within 2mm of sample edge for evaluation. Below blue area is evaluation area.



Martindale Abrasion Sample

B. Wet abrasion test

1. This test is run in a similar manner as the dry abrasion test except that the wool fabric abradant and the felt backer are soaked in distilled water before the test and re-wetted during the test. It is suggested that this test is started early in the working day so that it can be completed on the same day.
2. Soak the wool fabric abradant and felt by submerging it in room temperature water for 5-10 minutes. Ensure that the wool fabric and abradant are fully submerge in the water.
3. While these materials are soaking in the water, insert the test sample in the sample holder as described in **STEP 1** of the **DRY abrasion test**.
4. After 5-10minutes of soaking, remove the wool fabric abradant and felt backer from the water, allow excess water to drip off the fabric abradant and felt.
5. Position the fabric abradant and felt backer on the test station as described in "**Equipment Preparation**". (Please note that the placement of the 2.5kg weight will result in the "squeezing out of water", This water must be completely wiped off the machine.)
6. As quickly as possible, complete **Steps 2** and **3** as described earlier in the **Dry abrasion test** and begin the test.
7. Examine the samples at least every 2,000 cycles.
8. After 6,500 cycles stop the test and re-wet the fabric abradant and felt backer by gradually pouring up to 30 grams of distilled water onto the clamped fabrics, while lightly rubbing it in with the finger tips, stop pouring when no more water is being absorbed, Place the 2.5kg disc weight on the fabric and leave it in place for about 5 seconds to squeeze out excess water. Wipe off any excess water from the machine.
9. Re-start the test and examine at least every 2,000 cycles as before.
10. Discontinue the test according to the criteria described in **STEP 5** of the **Dry abrasion test.**
11. The Wet abrasion test should be run for a maximum of 13,000 cycles. If no damage has occurred at this point, discontinue the testing, and record the result as "13,000+ cycles".

Note: If the test cannot be completed on the same day as when it was started, the fabric abradant and felt backer must be removed from the machine and re-soaked in water as described in **STEP 2**.

REPORT: The lower number of cycles which were required to produce failure of a test sample.